

Step-15 Newton formula

$$f(x) > 0 \Leftrightarrow x = g(x)$$

$$g(x) = x - \frac{f(x)}{f'(x)}$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$f'(u, su) = \lim_{\epsilon \rightarrow 0} \frac{F(u + \epsilon su) - F(u)}{\epsilon} = \frac{G(u, su)}{\epsilon}$$

$$\text{Let } p = \nabla u, \quad q = \nabla su$$

$$\text{Then } G(u, su) = - \nabla \left(\frac{p + \epsilon q}{\sqrt{1 + |p + \epsilon q|^2}} \right) + \nabla \cdot \left(\frac{p}{\sqrt{1 + |p|^2}} \right)$$

$$* \quad 1 + |p + \epsilon q|^2 = \underbrace{1 + |p|^2}_A + \underbrace{\epsilon(2p \cdot q)}_B + \underbrace{O(\epsilon^2)}_C$$

(last term we don't consider as once we divide by ϵ and $\epsilon \rightarrow 0$ it will go to zero)

$$\left\{ \begin{aligned} f(x+h) &= f(x) + h f'(x) + o(h^2) \end{aligned} \right.$$

$$\text{choose } f(x) = \sqrt{x} \Rightarrow \sqrt{A + \epsilon B} = \sqrt{A} + \epsilon B \cdot \frac{1}{2\sqrt{A}} + o(\epsilon^2)$$

$$* \quad \sqrt{1 + |p + \epsilon q|^2} = \sqrt{A + \epsilon B} = \sqrt{A} + \frac{\epsilon B}{2\sqrt{A}} + o(\epsilon^2)$$

$$\text{choose } C = \frac{p \cdot q}{\sqrt{A}}$$

$$* \quad \frac{1}{\sqrt{A + \epsilon B}} = \frac{1}{\sqrt{A} + \epsilon C} = \frac{1}{\sqrt{A} \left(1 + \frac{\epsilon C}{\sqrt{A}} \right)}$$

$$\text{using } (1+x)^{-1} = 1 - x + o(x^2)$$

$$* \quad \frac{1}{\sqrt{A} \left(1 + \frac{\epsilon C}{\sqrt{A}} \right)} = \frac{1}{\sqrt{A}} \left(1 - \frac{\epsilon C}{\sqrt{A}} \right) + o(\epsilon^2)$$

$$\text{Finally, } \frac{1}{\sqrt{1 + |p + \epsilon q|^2}} = \frac{1}{\sqrt{1 + |p|^2}} - \frac{\epsilon p \cdot q}{(1 + |p|^2)^{3/2}} + o(\epsilon^2)$$

$$\Rightarrow \frac{G(u, s_u)}{\epsilon} = \frac{1}{\epsilon} \left(-\nabla \cdot (p + \epsilon q) \left[\frac{1}{\sqrt{1+|p|^2}} - \frac{\epsilon p \cdot q}{(1+|p|^2)^{3/2}} \right] + \nabla \cdot \left(\frac{p}{\sqrt{1+|p|^2}} \right) \right)$$

zero order term in ϵ cancel out, $\frac{\epsilon^2}{\epsilon}$ and taking $\epsilon \rightarrow 0$, this term will be zero. so finally (only ϵ terms left)

$$F'(u, s_u) = -\nabla \cdot \left(\frac{\nabla s_u}{\sqrt{1+|u|^2}} - \frac{(\nabla u \cdot \nabla s_u) \nabla u}{(1+|u|^2)^{3/2}} \right)$$